

MATERIAL SAFETY DATA SHEET

1. PRODUCT AND MANUFACTURER IDENTIFICATION

DIFENOCONAZOLE 150G/L + PROPICONAZOLE 150G/L EC

SUPPLIER NAME:	Shandong Weifang Rainbow Chemical Co., Ltd.
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CHEMICAL NAME	Difenoconazole: 3-chloro-4-[(2RS,4RS;2RS,4SR)-4-methyl-2-(1H-1,2,4-triazol-1-ylmethyl)-1,3-dioxolan-2-yl]phenyl 4-chlorophenyl ether Propiconazole: 1-[[2-(2,4-dichlorophenyl)-4-propyl-1,3-dioxolan-2-yl]-methyl]-1H-1,2,4-triazole.
CAS NUMBER	Difenoconazole: 119446-68-3 Propiconazole: 60207-90-1
MOLECULAR FORMULA	Difenoconazole: C ₁₉ H ₁₇ Cl ₂ N ₃ O ₃ Propiconazole: C ₁₅ H ₁₇ Cl ₂ N ₃ O ₂
MOLECULAR WEIGHT	Difenoconazole: 406.3 Propiconazole: 342.2
CHEMICAL FAMILY:	Triazole
USE:	Fungicide

2. COMPOSITION / INFORMATION ON INGREDIENTS

DIFENOCONAZOLE	150G/L
PROPICONAZOLE:	150G/L
OTHER INGREDIENTS	Up to 1 L

3. HAZARDS IDENTIFICATION

PRIMARY ROUTES OF EXPOSURE: The substance can be absorbed into the body by ingestion.

INHALATION: Not irritating

EYE CONTACT: Slightly irritating to eyes.

SKIN CONTACT: Not irritating

DELAYED EFFECTS: Not found

4. FIRST - AID MEASURES

INGESTION: If swallowed, seek medical advice immediately and show this container or label. Do not induce vomiting.

EYE CONTACT: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses. Immediate medical attention is required.

SKIN CONTACT: Take off all contaminated clothing immediately. Wash off immediately with plenty of water. If skin irritation persists, call a physician. Wash contaminated clothing before re-use.

INHALATION: Remove to fresh air. If breathing is irregular or stopped, administer artificial respiration. Keep patient warm and at rest. Call a physician or poison information center immediately.

NOTE TO PHYSICIAN: There is no specific antidote. Treat symptomatically.

5. FIRE FIGHTING MEASURES

SUITABLE EXTINGUISHING MEDIA: Foam, CO₂, and water.

FLAMMABILITY: Combustible in case of fire.

SPECIAL FIRE FIGHTING PROCEDURES: Wear approved self-contained breathing apparatus and full protective clothing. Contain water run-off.

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6. ACCIDENTAL RELEASE MEASURES

ENVIRONMENTAL PRECAUTIONS: Prevent further leakage or spillage if safe to do so. Do not flush into surface water or sanitary sewer system.

METHODS FOR CLEANING UP: Contain and collect spillage with non-combustible absorbent material, (e.g. sand, earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

ADDITIONAL ADVICE: If the product contaminates rivers and lakes or drains inform respective authorities.

7. HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. do not store food, beverages or tobacco products in the storage area. prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. wash thoroughly with soap and water after handling.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Protective measures: The use of technical measures should always have priority over the use of personal protective equipment.

When selecting personal protective equipment, seek appropriate professional advice. Personal protective equipment should be certified to appropriate standards.

Respiratory protection: A combination gas, vapor and particulate respirator may be necessary until effective technical measures are installed.

Protection provided by air-purifying respirators is limited. Use a self-contained breathing apparatus in cases of emergency spills, when exposure levels are unknown, or under any circumstances where air-purifying respirators may not provide adequate protection.

Hand protection: Chemical resistant gloves are not usually required. Select gloves based on the physical job requirements.

Eye protection: If eye contact is possible, use tight-fitting chemical safety goggles.

Skin and body protection: No special protective equipment required. Select skin and body protection based on the physical job requirements.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE:	Yellow to brown liquid
ODOR	Characteristic odor
SPECIFIC GRAVITY	About 1.098 (at 20°C)
pH:	5 - 8 at 1 % w/v
OXIDIZING PROPERTIES:	Not oxidizing
EXPLOSIVE PROPERTIES:	Explosive properties
SOLUBILITY IN WATER:	Form emulsion with water
Boiling point	Not available

10. STABILITY AND REACTIVITY

Hazardous decomposition products: Combustion or thermal decomposition will evolve toxic and irritant vapors.

Hazardous reactions: None known.

Hazardous polymerization does not occur.

Stable under normal conditions.

11. TOXICOLOGICAL INFORMATION

DIFENOCONAZOLE:

ACUTE ORAL TOXICITY:	LD50: > 4000 mg/kg in rats
ACUTE DERMAL TOXICITY:	LD50: > 4000 mg/kg in rats
INHALATION:	LC50 (4h) in rats was > 2.90 mg/L air in rats
ACUTE SKIN IRRITATION:	Slight-irritant (rabbit)
ACUTE EYE IRRITATION:	Slight - irritant (rabbit)

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SKIN SENSITIZATION: Not sensitizing (guinea pig)

12. ECOLOGICAL INFORMATION

ELIMINATION INFORMATION (PERSISTENCE AND DEGRADABILITY)

Bioaccumulation:

Difenoconazole has high potential to bioaccumulate.

Propiconazole has low to medium potential to bioaccumulate.

Stability in water:

Difenoconazole is persistent in water

Propiconazole is stable in water

Stability in soil:

Difenoconazole is not persistent in soil.

Propiconazole is not persistent in soil.

Mobility:

Difenoconazole has low mobility in soil.

Propiconazole has low to medium mobility in soil.

ECOTOXICITY EFFECTS

Toxicity to fish:

LC50 *Oncorhynchus mykiss* (rainbow trout), 4.7 mg/l, 96 h

Based on test results obtained with similar product.

Toxicity to aquatic invertebrates: EC50 *Daphnia magna* Straus, 2.7 mg/l, 48 h

Based on test results obtained with similar product.

Toxicity to algae: EBC50 *scenedesmus subspicatus*, 0.23 mg/l, 72 h

Based on test results obtained with similar product.

13. DISPOSAL CONSIDERATIONS

Single rinse before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemicals on site. Puncture or shred and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

14. TRANSPORT INFORMATION

UN-No: 3082

Class: 9

Danger Label Number: 9

Packaging group III

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

(PROPICONAZOLE AND DIFENOCONAZOLE)

15. OTHER INFORMATION

The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety requirements. The data does not signify any warranty with regard to the products' properties.